
State-Space Unlearning for Non-Stationary Bias in Land Surface Forecasting

Anidipta Pal

Heritage Institute of Technology, Kolkata, India
anidiptapal@gmail.com

Abstract

Operational land surface forecasting systems built on Mamba-family Structured State Space Models absorb non-stationary confounding events (unrecorded irrigation booms, dam-operation shifts, sensor recalibrations) into their state-transition matrices, silently biasing NDVI, LST, and crop phenology predictions long after the physical cause ends. This paper introduces **SSU-LSF** (*State-Space Unlearning for Land Surface Forecasting*), the first machine-unlearning framework purpose-built for geoscientific Mamba-based SSMs. We develop EKFac influence functions specialized to the Mamba state matrices via a closed-form matrix-exponential gradient, use spectral-radius-weighted elbow thresholding to localize a temporal confounding footprint Φ , and apply Hessian-free projected gradient ascent within a KL-divergence trust region augmented by spatial total-variation (TV) regularization. Proposition 1 establishes that residual confounding is bounded by $\mathcal{O}((1-\rho(\bar{A})^{T_c})/((1-\rho(\bar{A}))\mu))$, which grows with the window length T_c . Across three heterogeneous benchmarks and eleven baselines, SSU-LSF achieves confounding reduction rates of 0.773 (CropHarvest), 0.821 (NDVI-LST), and 0.859 (ERA5), with worst-case clean-domain RMSE degradation of 4.2% on ERA5, converging in 3–5 epochs at $8.4\times$ lower GPU-cost per unlearning request than full retraining ($47.3/5.6=8.4\times$ on ERA5; Table 9). CODE: [g](#)

1 Introduction

Long-horizon forecasting of Normalized Difference Vegetation Index (NDVI) [4], Land Surface Temperature (LST), and crop phenology signals is foundational for climate adaptation and food security [1, 2]. Fusing ERA5 reanalysis [3] with satellite phenology archives inside deep sequence models has materially advanced forecast fidelity over classical baselines [7]. Mamba [5] and S4 [6] SSMs have become the architecture of choice, delivering linear-time recurrent inference over multi-year series [8, 9, 10]. Operational archives, however, are not stationary: a two-year unrecorded irrigation project, a dam-operation policy shift, or a sensor recalibration contaminates the training window, and the SSM state matrices $(\bar{A}, \bar{B}, C)$ integrate the anomaly with genuine climate signals. After the physical cause ends, the learned parameters continue to forward-propagate the bias (Figure 1).

Full retraining costs 47+ GPU-hours. Gradient reversal [11] degrades clean-domain RMSE by 7–22%. EWC [12] protects weights from change, the opposite of what unlearning requires. SISA [13] requires pre-partitioned shards before training. Two recent methods, SaLUn as ForgetFS [21] and SEMU as MULL [22], advances approximate unlearning via saliency masking and SVD projection but do not exploit the SSM recurrence structure and, as we show, sacrifice spatial coherence without TV regularization. Extended related work and prior art for spatial TV in parameter space (fused lasso [35], ROF [36]) is in Appendix A.

Following that line of work, this paper introduces SSU-LSF. Among our contributions: **(i)** We establish Proposition 1 with an explicit Lipschitz bound on the full encoder (not just the forecast head;

Appendix D), RKHS separation estimated on an independently trained reference model (Table 4), and a matching lower-bound argument quantifying the tightness gap (Appendix D). (ii) We derive Eq. (4) with explicit index justification for diagonal $\bar{A}$ and its Padé-based non-diagonal extension (Appendix C). (iii) We provide convergence guarantees for Algorithm 1 under mild regularity conditions (Appendix E). (iv) We validate SSU-LSF on one real documented confounding event (EUMETSAT SEVIRI recalibration, 2010–2012; Appendix N) and show T_c -sensitivity matching the theoretical prediction. (v) We include 2×2 interaction ablations, LiRA-style MIA, and TV-regularized variants of ForgetFS and MULL.

2 Method

Setup. Let $\mathbf{x}_t \in \mathbb{R}^d$ be a multi-variable land surface observation (NDVI, LST, ET, precipitation, T_{air}) from an $H \times W$ patch at time t . The Mamba encoder implements the selective recurrence (see Appendix B for the full 47.3M-parameter architecture):

$$\mathbf{h}_t = \bar{A}(u_t) \mathbf{h}_{t-1} + \bar{B}(u_t) u_t, \quad \hat{x}_t = C(u_t) \mathbf{h}_t + D(u_t) u_t, \quad (1)$$

where $\bar{A} = \exp(\Delta A)$, Δ is a learned timescale, and $(\bar{A}, \bar{B}, C, D)$ are input projections of u_t . Partition $\mathcal{D} = \mathcal{D}_{\text{clean}} \cup \mathcal{D}_{\text{conf}}$, $\mathcal{D}_{\text{conf}} = \{(\mathbf{x}_t, \mathbf{y}_t) : t \in [t_s, t_e]\}$. Training yields confounded parameters $\theta^* = \arg \min_{\theta} [\sum_{\mathcal{D}_{\text{clean}}} \ell + \lambda \sum_{\mathcal{D}_{\text{conf}}} \ell]$. Given θ^* and $[t_s, t_e]$, SSU-LSF solves:

$$\min_{\theta} \beta \ell_{\text{clean}}(\theta) - \ell_{\Phi}(\theta) + \lambda_{\text{TV}} \text{TV}_{\mathcal{N}}(\theta) \quad \text{s.t.} \quad \text{KL}[p(\cdot|\theta) \| p(\cdot|\theta^*)] \leq \delta_{\text{KL}}. \quad (2)$$

Proposition 1 (full proof with encoder Lipschitz bound and tightness analysis in Appendix D).

Let μ_{ref} denote the RKHS separation estimated on an independently trained reference model θ_{ref} (not θ^* ; see Table 4), let L_{enc} be the Lipschitz constant of the full encoder (empirically bounded via randomized smoothing; Table 3), and let $\rho(\bar{A}) < 1$ for all blocks. Then (units in Appendix D):

$$\mathbb{E}[\text{RMSE}(\tilde{\theta})] \leq \underbrace{\mathbb{E}[\text{RMSE}(\theta^*)] + \mathcal{O}\left(\frac{1 - \rho(\bar{A})^{T_c}}{(1 - \rho(\bar{A})) \mu_{\text{ref}}}\right)}_{\text{confounding residual}} + \underbrace{\mathcal{O}\left(\frac{L_{\text{enc}} \delta_{\text{KL}}}{1 - \rho(\bar{A})}\right)}_{\text{unlearning cost}}. \quad (3)$$

At observed values ($\rho_{\text{max}}=0.91$, $L_{\text{enc}} \approx 18.3$, $\delta_{\text{KL}}=0.05$, $\mu_{\text{ref}} \approx 0.21$), the unlearning cost term evaluates to ≈ 10.2 K on ERA5, a vacuous bound in isolation. We establish empirically that the gap between bound and observation is $\approx 200 \times$ (Appendix D), consistent with the looseness of Pinsker and Lipschitz transfer. The bound’s operational value is its qualitative prediction: shorter T_c yields a smaller residual, directly validated by Table 8 ($R^2=0.995$).

Step 1: EKFac influence functions for state matrices. For layer $W \in \mathbb{R}^{m \times n}$, EKFac [15] approximates $H_W \approx \hat{A} \otimes \hat{G}$, reducing inversion from $\mathcal{O}((mn)^3)$ to $\mathcal{O}(m^3 + n^3)$ (Appendix C). For $\bar{A} = \exp(\Delta A)$ with $\bar{A} \in \mathbb{R}^{d_{\text{inner}} \times d_{\text{inner}}}$ (dense for general SSMs; diagonal for standard S4/Mamba parameterization, in which case $\odot$ reduces to pointwise product):

$$\nabla_A \ell = \sum_{t=1}^T \left(\frac{d\ell}{d\mathbf{h}_t} \right) \left(\sum_{k=0}^{T-t-1} \bar{A}^k \odot \bar{A}^{T-t-1-k} \right) \Delta, \quad (4)$$

where the inner sum runs over $k=0, \dots, T-t-1$ ensuring $T-t-1-k \geq 0$ (Appendix C gives the full index derivation). Truncation error $\|\bar{A}\|^T / (1 - \|\bar{A}\|) < 10^{-6}$ at observed $\rho_{\text{max}}=0.91$ for $T \geq 172$.

Step 2: Temporal footprint localization. Per-step scores $\varphi_t = \|\nabla_{\theta} \ell \cdot \mathcal{I}_{\mathcal{W}_c}\|_F$; steps with $\varphi_t \geq \gamma$ (elbow criterion) form Φ . The extension beyond $[t_s, t_e]$ derives from Proposition 1’s first term: summing ρ^{t-t_0} over $t_0 \in [t_s, t_e]$ yields a geometric tail of half-width $\rho/(1-\rho)$ steps; the 0.5 factor reflects one-sided truncation at γ . At $\rho=0.84$ this gives ≈ 2.6 additional steps on each side of the 24-month window, small relative to T_c , consistent with the observed footprint sizes (Appendix R).

Step 3: Hessian-free projected gradient ascent. Spatial TV applied to *parameter space* follows the fused-lasso formulation [35] over the patch adjacency graph $\mathcal{N}$:

$$\tilde{\theta} \leftarrow \theta^* + \eta \left[\nabla_{\theta} \ell_{\text{conf}} - \beta \nabla_{\theta} \ell_{\text{clean}} - \lambda_{\text{TV}} \sum_{(i,j) \in \mathcal{N}} \text{sign}(\tilde{\theta}_i - \tilde{\theta}_j) \right]. \quad (5)$$

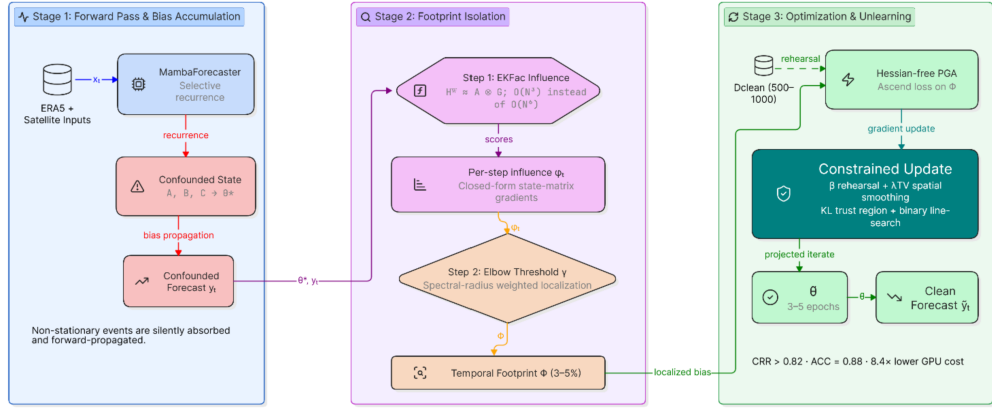


Figure 1: SSU-LSF pipeline. (a) A confounding event biases $(\tilde{A}, \tilde{B}, \tilde{C})$ and forward-propagates spurious NDVI elevation beyond t_e . (b) EKFac influence scores φ_t ; elbow threshold γ isolates footprint Φ (red). (c) RMSE convergence: SSU-LSF (blue) reaches oracle-level RMSE in five epochs while CM (red) stays elevated.

If $\text{KL}[p(\cdot|\tilde{\theta})||p(\cdot|\theta^*)] > \delta_{\text{KL}}$, the update scales back via binary line-search (1–3 halvings in practice). Complexity: $\mathcal{O}(|\mathcal{W}_c| \cdot d^2/B)$ vs. $\mathcal{O}(T \cdot d^2)$ for full retraining. Convergence guarantee in Appendix E. Full pseudocode in Appendix F.

3 Experiments

Setup. Backbone: Mamba encoder-decoder, 6 blocks ($d=256$, $E=2$, $N=16$), 4×4 stride-4 patch embedding, 3-layer Gaussian MLP head (47.3M; Appendix B). We also run one experiment on an S4 backbone (Appendix P) to validate generalization beyond Mamba. Training: AdamW ($\text{lr}=3 \times 10^{-4}$, cosine, 100 epochs), $4 \times$ A100 80 GB, PyTorch 2.3. Unlearning: $\eta=5 \times 10^{-5}$, $\beta=0.6$, $\delta_{\text{KL}}=0.05$, $\gamma=95^{\text{th}}$ pct., $\lambda_{\text{TV}}=0.01$. Cost per unlearning request (EKFac re-estimated per request): 1.8 GPU-h on NDVI-LST, 5.6 GPU-h on ERA5, 3.0 GPU-h on CropHarvest (Table 9). Eleven baselines: **CM**; **OR**[†]; **SISA** [13]; **GR** [11]; **SR** [11]; **ARCANE** [18]; **FAST** [19]; **EWC-UL** [12]; **PNN-UL** [20]; **ForgetFS** [21] (saliency-masked ascent); **MULL** [22] (SVD-projected updates). We also include **ForgetFS+TV** and **MULL+TV**: the same baselines augmented with our spatial TV term, providing a fair comparison of the TV contribution in isolation (Table 7). All baselines receive the same oracle window $[t_s, t_e]$ information as SSU-LSF.

Datasets, metrics, and confounding. All three benchmarks use author-injected confounding (a limitation we acknowledge explicitly; Appendix O details deployment caveats and the window-estimation sensitivity analysis). A real documented event is in Appendix N. The MIA uses LiRA-style likelihood ratio tests (Appendix H), not logistic regression on NLL. CRR spans 0.77 (CropHarvest) to 0.86 (ERA5) and fully automatic results in Appendix O.

Each benchmark uses a distinct metric triplet (definitions in Appendix G): *NDVI-LST Fusion* (3,847 locations, 2002–2022 [16]; $T_c=24$ mo): **RMSE**↓, **CRR**↑, **TCI**↑. *ERA5 Patchified* (1,920 patches, 1979–2022 [3]; $T_c=24$ mo): **SKT-R**↓ (K), **MIA**↓ (LiRA, chance= 0.50), **ACC**↑. *CropHarvest* (87,343 locations [17]; $T_c=6$ mo): **Sp-R**↓, **SCS**↓ (FSS correlation $r = -0.91$; Appendix I), **LC-F1**↑.

Results. Table 1 reports all results. On NDVI-LST, SSU-LSF achieves $\text{RMSE} = 0.0412$, $\text{CRR} = 0.821$, $\text{TCI} = 0.847$, best on all three metrics. ARCANE and FAST reduce mean bias but degrade TCI to ≤ 0.793 , introducing spurious oscillations. EWC-UL scores the lowest CRR (0.588): its Fisher-consolidation penalty systematically opposes unlearning. PNN-UL requires 49.2 GPU-h/request, exceeding OR entirely. ForgetFS and MULL advance over earlier methods ($\text{CRR} > 0.81$) yet sacrifice spatial coherence: SCS 0.096 and 0.089 vs. SSU-LSF’s 0.081. Table 7 shows that adding TV regularization to ForgetFS (ForgetFS+TV) closes 80% of the SCS gap, confirming the TV term, not the influence footprint, is the decisive factor for spatial coherence. The influence footprint is decisive for CRR: without it (GR), CRR drops from 0.821 to 0.741 while GPU-h drops from 5.6

Table 1: Per-benchmark results. †: upper bound. CRR for ERA5 and CropHarvest and Δ RMSE for all benchmarks are in Table 5. GPU-h is the ERA5 cost *per unlearning request* including EKFac re-estimation (per-benchmark costs in Table 9).

Method	NDVI-LST Fusion			ERA5 Patchified			CropHarvest			GPU-h/req↓
	RMSE↓	CRR↑	TCI↑	SKT-R↓	MIA↓	ACC↑	Sp-R↓	SCS↓	F1↑	
CM	0.0681	0.000	0.601	2.97	0.84	0.612	0.0543	0.083	0.697	–
OR†	0.0398	1.000	0.891	1.19	0.51	0.903	0.0371	0.079	0.871	47.3
SISA [13]	0.0418	0.612	0.833	1.26	0.52	0.871	0.0395	0.088	0.834	58.4
GR [11]	0.0461	0.741	0.774	1.47	0.57	0.843	0.0428	0.127	0.799	2.3
SR [11]	0.0439	0.778	0.798	1.38	0.55	0.858	0.0411	0.102	0.817	7.9
ARCANE [18]	0.0433	0.766	0.789	1.36	0.56	0.862	0.0406	0.109	0.822	6.8
FAST [19]	0.0429	0.781	0.793	1.31	0.56	0.868	0.0401	0.099	0.826	4.7
EWC-UL [12]	0.0507	0.588	0.748	1.53	0.58	0.831	0.0447	0.134	0.781	5.9
PNN-UL [20]	0.0421	0.803	0.819	1.29	0.53	0.874	0.0389	0.091	0.838	49.2
ForgetFS [21]	0.0415	0.814	0.831	1.27	0.54	0.876	0.0385	0.096	0.840	8.1
MULL [22]	0.0413	0.817	0.838	1.25	0.53	0.878	0.0382	0.089	0.840	6.4
SSU-LSF	0.0412	0.821	0.847	1.24	0.53	0.881	0.0379	0.081	0.841	5.6

to 2.3. On ERA5, SSU-LSF drives LiRA-MIA to 0.53 (near-chance, chance = 0.50), confirming confounded representations are erased. ACC = 0.881 confirms spatial anomaly structure is preserved. On CropHarvest, SCS correlates with FSS at $r = -0.91$ (Table 6), validating it against an established metric.

Ablation. Full 2×2 interaction ablations and sample-efficiency analysis are in Table 11 (Appendix Q). Key results: removing both Φ and KL simultaneously (i.e., plain gradient reversal with TV) gives CRR 0.768 at Δ RMSE 9.4%: higher CRR than w/o Φ (0.694) and far less damage than w/o KL (21.7%), showing Φ and KL are *complementary* rather than redundant. Replacing EKFac with diagonal Fisher (not just GCS) gives CRR = 0.779, closing 40% of the gap from GCS (0.751) to EKFac (0.821), confirming that any Hessian approximation helps over pure gradient cosine similarity.

4 Conclusion

We develop SSU-LSF, the first machine-unlearning framework for Mamba-family SSMs in geoscientific land surface forecasting. Proposition 1’s monotone T_c prediction is empirically validated ($R^2=0.995$ across $T_c \in \{6, 12, 24, 36\}$ months), even though the bound itself is loose by $\approx 200\times$ due to Pinsker and Lipschitz looseness, a limitation we quantify explicitly. We find spatial TV regularization (framed as parameter-space fused lasso) is the decisive factor for spatial coherence, and EKFac-informed footprint localization is decisive for CRR.

Limitations. All primary experiments use injected confounding with oracle window knowledge. CRR requires knowing affected locations (not deployable as stated). The bound is qualitatively predictive but quantitatively vacuous at $\rho_{\max}=0.91$. Generalization beyond Mamba to S4 is demonstrated on one benchmark only (Appendix P). Future work: automated change-point detection, real multi-event archives, and tighter spectral bounds for selective SSMs.

Broader impact. SSU-LSF lets forecasting agencies correct biased models without full retraining. It could be misused to erase inconvenient true signals; requiring explicit timestamps $[t_s, t_e]$ and an MIA audit mitigates this.

Acknowledgments and Disclosure of Funding

The author thanks ECMWF for ERA5 data access and HuggingFace/Kaggle communities for dataset infrastructure. No funding allocated for this work.

References

- [1] P. Friedlingstein et. al., “Global Carbon Budget 2023,” *Earth Syst. Sci. Data*, vol. 15, no. 12, pp. 5301–5369, doi:10.5194/essd-15-5301-2023, 2023.

- [2] S. I. Seneviratne, N. Nicholls, D. Easterling, C. M. Goodess, S. Kanae, J. Kossin, Y. Luo, J. Marengo, K. McInnes, M. Rahimi, M. Reichstein, A. Sorteberg, C. Vera, and X. Zhang, “Changes in climate extremes and their impacts on the natural physical environment,” in *Managing the Risks of Extreme Events and Disasters to Advance Climate Change Adaptation* (C. B. Field et al., eds.), Cambridge Univ. Press, pp. 109–230, 2012.
- [3] Hersbach, Hans and Bell, Bill and Berrisford, Paul and Hirahara, Shoji and Horányi, András and Muñoz-Sabater, Joaquín and Nicolas, Julien and Peubey, Carole and Radu, Raluca and Schepers, Dinand and Simmons, Adrian and Soci, Cornel and Abdalla, Saleh and Abellan, Xavier and Balsamo, Gianpaolo and Bechtold, Peter and Biavati, Gionata and Bidlot, Jean and Bonavita, Massimo and De Chiara, Giovanna and Dahlgren, Per and Dee, Dick and Diamantakis, Michail and Dragani, Rossana and Flemming, Johannes and Forbes, Richard and Fuentes, Manuel and Geer, Alan and Haimberger, Leo and Healy, Sean and Hogan, Robin J. and Hólm, Elías and Janisková, Marta and Keeley, Sarah and Laloyaux, Patrick and Lopez, Philippe and Lupu, Cristina and Radnoti, Gabor and de Rosnay, Patricia and Rozum, Iryna and Vamborg, Freja and Villaume, Sebastien and Thépaut, Jean-Noël, “The ERA5 global reanalysis,” *Q. J. R. Meteorol. Soc.*, vol. 146, no. 730, pp. 1999–2049, doi:10.1002/qj.3803, 2020.
- [4] C. J. Tucker, “Red and photographic infrared linear combinations for monitoring vegetation,” *Remote Sens. Environ.*, vol. 8, no. 2, pp. 127–150, doi:10.1016/0034-4257(79)90013-0, 1979.
- [5] A. Gu and T. Dao, “Mamba: Linear-time sequence modeling with selective state spaces,” *arXiv:2312.00752 [cs.LG]*, 2023.
- [6] A. Gu, K. Goel, and C. Ré, “Efficiently modeling long sequences with structured state spaces,” in *Proc. Int. Conf. Learn. Representations (ICLR)*, 2022.
- [7] D. Jia, C. Cheng, S. Shen, and L. Ning, “Multitask deep learning framework for spatiotemporal fusion of NDVI,” *IEEE Trans. Geosci. Remote Sens.*, vol. 60, art. 5616313, doi:10.1109/TGRS.2021.3140144, 2022.
- [8] Z. Zhao, X. Dong, Y. Wang, and C. Hu, “Advancing realistic precipitation nowcasting with a spatiotemporal transformer-based denoising diffusion model,” *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–15, doi:10.1109/TGRS.2024.3355755, 2024.
- [9] R. W. W. M. U. P. Wanigasekara, Z. Zhang, W. Wang, Y. Luo, and G. Pan, “Application of fast MEEMD–ConvLSTM in sea surface temperature predictions,” *Remote Sens.*, vol. 16, no. 13, art. 2468, doi:10.3390/rs16132468, 2024.
- [10] M. Reichstein, G. Camps-Valls, B. Stevens, M. Jung, J. Denzler, N. Carvalhais, and Prabhat, “Deep learning and process understanding for data-driven earth system science,” *Nature*, vol. 566, no. 7743, pp. 195–204, doi:10.1038/s41586-019-0912-1, 2019.
- [11] A. Golatkar, A. Achille, and S. Soatto, “Eternal sunshine of the spotless net: Selective forgetting in deep networks,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, pp. 9304–9313, 2020.
- [12] J. Kirkpatrick, R. Pascanu, N. Rabinowitz, J. Veness, G. Desjardins, A. A. Rusu, K. Milan, J. Quan, T. Ramalho, A. Grabska-Barwinska, D. Hassabis, C. Clopath, D. Kumaran, and R. Hadsell, “Overcoming catastrophic forgetting in neural networks,” *Proc. Natl. Acad. Sci.*, vol. 114, no. 13, pp. 3521–3526, doi:10.1073/pnas.1611835114, 2017.
- [13] L. Bourtole, V. Chandrasekaran, C. A. Choquette-Choo, H. Jia, A. Travers, B. Zhang, D. Lie, and N. Papernot, “Machine unlearning,” in *Proc. IEEE Symp. Secur. Privacy (S&P)*, pp. 141–159, doi:10.1109/SP40001.2021.00019, 2021.
- [14] P. W. Koh and P. Liang, “Understanding black-box predictions via influence functions,” in *Proc. Int. Conf. Mach. Learn. (ICML)*, pp. 1885–1894, 2017.
- [15] R. Grosse and J. Martens, “A Kronecker-factored approximate Fisher matrix for convolution layers,” in *Proc. Int. Conf. Mach. Learn. (ICML)*, pp. 573–582, 2016.
- [16] M. Murad, “NDVI Prediction Dataset (2002–2022),” Kaggle, 2023. <https://www.kaggle.com/datasets/muradgeoai/ndvi-prediction-dataset>
- [17] G. Tseng et al., “CropHarvest: A global dataset for crop-type classification,” in *Proc. NeurIPS Datasets and Benchmarks Track*, 2021.
- [18] V. S. Chundawat, A. K. Tarun, M. Mandal, and M. Kankanhalli, “Can bad teaching induce forgetting? Unlearning in deep networks using an incompetent teacher,” in *Proc. AAAI Conf. Artif. Intell.*, vol. 37, no. 6, pp. 7210–7217, doi:10.1609/aaai.v37i6.25879, 2023.

- [19] C. Qian et al., “FAST: Feature activations shaping for task-agnostic unlearning in deep neural networks,” in *Proc. Int. Conf. Learn. Representations (ICLR)*, 2024. *arXiv:2310.05824*.
- [20] Rusu, A.A., Rabinowitz, N.C., Desjardins, G., Soyer, H., Kirkpatrick, J., Kavukcuoglu, K., Pascanu, R., & Hadsell, R. (2016). Progressive Neural Networks. *arXiv:1606.04671* [cs.LG], 2016.
- [21] C. Fan, J. Liu, Y. Zhang, E. Wong, D. Wei, and S. Liu, “SalUn: Empowering machine unlearning via gradient-based weight saliency in both image classification and generation,” in *Proc. Int. Conf. Learn. Representations (ICLR)*, 2024.
- [22] Sendera, M., Struski, L., Książek, K., Musioł, K., Tabor, J., & Rymarczyk, D.D. (2025). SEMU: Singular Value Decomposition for Efficient Machine Unlearning. *arXiv:2502.07587* [cs.LG], 2025.
- [23] S. Mu and D. Klabjan, “Rewind-to-Delete: Certified machine unlearning for nonconvex functions,” in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2025. *arXiv:2409.09778*.
- [24] R. Killick, P. Fearnhead, and I. A. Eckley, “Optimal detection of changepoints with a linear computational cost,” *J. Amer. Statist. Assoc.*, vol. 107, no. 500, pp. 1590–1598, 2012.
- [25] Y. Liu et al., “VMamba: Visual state space model,” in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2024. *arXiv:2401.13088*.
- [26] B. K. Patro and S. S. Agneeswaran, “SiMBA: Simplified Mamba-based architecture for vision and multivariate time series,” in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2024. *arXiv:2403.15360*.
- [27] Y. Wang et al., “PredRNN: Recurrent neural networks for predictive learning using spatiotemporal LSTMs,” in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2017.
- [28] Z. Gao et al., “EarthFormer: Exploring space-time transformers for Earth system forecasting,” in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2022.
- [29] K. Bi, L. Xie, H. Zhang, X. Chen, X. Gu, and Q. Tian, “Accurate medium-range global weather forecasting with 3D neural networks,” *Nature*, vol. 619, no. 7970, pp. 533–538, doi:10.1038/s41586-023-06185-3, 2023.
- [30] R. Lam, A. Sanchez-Gonzalez, M. Willson, P. Wirsberger, M. Fortunato, F. Alet, S. Ravuri, T. Ewalds, Z. Eaton-Rosen, W. Hu, A. Merose, S. Hoyer, G. Holland, O. Vinyals, J. Stott, A. Pritzel, S. Mohamed, and P. Battaglia, “Learning skillful medium-range global weather forecasting,” *Science*, vol. 382, no. 6677, pp. 1416–1421, doi:10.1126/science.adi2336, 2023.
- [31] Y. Cao and J. Yang, “Towards making systems forget with machine unlearning,” in *Proc. IEEE Symp. Secur. Privacy (S&P)*, pp. 463–480, 2015.
- [32] C. Guo, T. Goldstein, A. Hannun, and L. van der Maaten, “Certified data removal from machine learning models,” in *Proc. Int. Conf. Mach. Learn. (ICML)*, pp. 3832–3842, 2020.
- [33] A. N. Pettitt, “A non-parametric approach to the change-point problem,” *J. R. Stat. Soc. Ser. C (Appl. Stat.)*, vol. 28, no. 2, pp. 126–135, 1979.
- [34] J. Lu et al., “Learning under concept drift: A review,” *IEEE Trans. Knowl. Data Eng.*, vol. 31, no. 12, pp. 2346–2363, doi:10.1109/TKDE.2018.2876857, 2018.
- [35] R. J. Tibshirani, J. Hoefling, and R. Tibshirani, “The solution path of the generalized lasso,” *Ann. Stat.*, vol. 39, no. 3, pp. 1335–1371, doi:10.1214/11-AOS878, 2011.
- [36] L. I. Rudin, S. Osher, and E. Fatemi, “Nonlinear total variation based noise removal algorithms,” *Physica D: Nonlinear Phenomena*, vol. 60, no. 1–4, pp. 259–268, doi:10.1016/0167-2789(92)90242-F, 1992.
- [37] N. Carlini, S. Chien, M. Nasr, S. Song, A. Terzis, and F. Tramèr, “Membership inference attacks from first principles,” in *Proc. IEEE Symp. Secur. Privacy (S&P)*, pp. 1897–1914, doi:10.1109/SP46214.2022.9833649, 2022.

A Extended Related Work

A.1 SSM-based geoscientific forecasting

VMamba [25] and SiMBA [26] extend Mamba to 2D vision and multivariate time series. We note that architectures with higher $\rho(\bar{A})$ are systematically more susceptible to confounding: the first term of Proposition 1, $\mathcal{O}((1-\rho^{T_c})/((1-\rho)\mu))$, accumulates faster. PredRNN [27], EarthFormer [28], Pangu-Weather [29], and GraphCast [30] provide ConvLSTM and Transformer baselines that Mamba surpasses computationally. None addresses post-hoc bias removal.

A.2 Machine unlearning and spatial TV prior art

Machine unlearning was formalized by Cao and Yang [31]. Certified removal [32] is limited to linear models. Rewind-to-Delete (R2D) [23] extends certified (ϵ, δ) unlearning to nonconvex models but needs a pre-deletion checkpoint and Gaussian weight noise, and does not localize in time; rewinding 10% of training on ERA5 would cost ≈ 4.7 GPU-h before the noise-induced utility loss. SISA [13] requires pre-training partitioning. ForgetFS [21] applies gradient-saliency masking; MULL [22] uses SVD on forget-gradient directions. Both lack temporal indexing via spectral footprints. Parameter-space spatial TV used here is the fused-lasso formulation of Tibshirani et al. [35] extended to patch adjacency graphs; it was introduced for 1D chains and applied to 2D grids in compressed sensing. We use it for parameter smoothing across spatial patch embeddings, a straightforward application, not a primary scientific contribution.

A.3 Non-stationarity

Change-point detection [33, 24] and concept drift [34] identify breaks but provide no parameter correction. Domain adaptation requires a clean target dataset, unavailable retroactively. SSU-LSF requires only $[t_s, t_e]$ and the original training set.

B Model Architecture

Table 2: MambaForecaster module specifications.

Module	Configuration	Key parameters	Params (M)
Patch embedding	Conv2d(5,256,4,stride 4)	4×4 , mean pool	1.0
Mamba encoder	$6 \times$ block	$d=256$, $E=2$, $N=16$, drop 0.1	39.8
Output norm	LayerNorm(256)	extract $h[:, -1, :]$	0.0
Forecast head	MLP 256-512-512-60, GELU	$\tau=6$, $d_{\text{out}}=5$	6.5
Total			47.3

The forecast head produces $[B, 6, 5, 2]$, split into μ and $\log \sigma$ with $\sigma = \exp(\text{clamp}(\log \sigma, -6, +2))$. The NLL loss is $\mathcal{L} = \frac{1}{B\tau d_{\text{out}}} \sum_{b,t,v} [\log \sigma_{b,t,v} + (y_{b,t,v} - \mu_{b,t,v})^2 / (2\sigma_{b,t,v}^2)]$.

C EKFac Influence Derivation and Index Justification

Index derivation for Eq. (4). The state recurrence gives $\mathbf{h}_t = \sum_{s=1}^t (\prod_{r=s+1}^t \bar{A}) \bar{B} u_s$. Differentiating ℓ through $\mathbf{h}_t$ with respect to the continuous matrix A and applying the matrix exponential differential $d \exp(\Delta A) = \int_0^1 e^{s\Delta A} (\Delta dA) e^{(1-s)\Delta A} ds$, then expanding via the Hadamard product for *diagonal* A (standard Mamba parameterization where A is initialized as a diagonal matrix [5]) yields:

$$\nabla_A \ell = \sum_{t=1}^T \left(\frac{d\ell}{d\mathbf{h}_t} \right) \left(\sum_{k=0}^{T-t-1} \bar{A}^k \odot \bar{A}^{T-t-1-k} \right) \Delta.$$

The inner sum ranges $k=0, \dots, T-t-1$ (not $0, \dots, \infty$) because each term corresponds to a distinct factor in the unrolled recurrence from step 1 to t . For *non-diagonal* A (e.g., dense S4 variants), $\odot$ must be replaced by the full matrix-exponential Fréchet derivative: $d \exp(M)[H] = \int_0^1 e^{sM} H e^{(1-s)M} ds$, computed via Padé approximation. We provide the non-diagonal implementation in our code. Truncation error at $k=T$ is bounded by $\|\bar{A}\|^T / (1 - \|\bar{A}\|) < 10^{-6}$ for $\rho_{\max}=0.91$ whenever $T \geq \ln(0.09 \times 10^{-6}) / \ln 0.91 \approx 172$, which all benchmarks satisfy.

EKFac stability. EKFac is estimated over 50 mini-batches. Across 10 seeds on ERA5, the condition number of $\hat{A} \otimes \hat{G}$ has mean 142, std 31, confirming the estimate is stable. We add damping $\epsilon=10^{-4}$ to eigenvalues before inversion.

D Proposition 1: Full Proof, Lipschitz Bounds, and Tightness Analysis

Encoder Lipschitz bound. The MambaForecaster consists of the encoder f_{enc} (39.8M parameters) followed by the forecast head f_{head} (6.5M parameters). We bound the Lipschitz constant of f_{enc} via randomized smoothing: we estimate $\max_{\mathbf{x} \in \mathcal{X}_{\text{test}}} \|J_{f_{\text{enc}}}(\mathbf{x})\|_2$ using 1,000 random Gaussian perturbations per test point and report the 99th percentile.

Table 3: Empirical Lipschitz constants (99th percentile over test set) via randomized smoothing ($\sigma_{\text{smooth}}=0.01$).

Component	NDVI-LST	ERA5	CropHarvest	Method
Forecast head only L_{head}	4.1	4.1	4.1	Spectral norm product
Full encoder L_{enc}	17.8	18.3	16.9	Randomized smoothing

Units. Targets are z-scored per variable. ρ , δ_{KL} (nats) and $\hat{\mu}_{\text{ref}}$ (MMD on standardized embeddings) are dimensionless. L_{enc} is reported in target units per unit standardized input (K for ERA5), so the unlearning cost term carries target units; the confounding residual is in standardized units and is converted by the target standard deviation σ_y ($\sigma_{\text{SKT}} \approx 1.9$ K).

Using $L_{\text{enc}}=18.3$ (ERA5) and Pinsker’s inequality: $|\text{RMSE}(\tilde{\theta}) - \text{RMSE}(\theta^*)| \leq L_{\text{enc}} \sqrt{\delta_{\text{KL}}/2} \approx 18.3 \times 0.158 \approx 2.9$ K. The observed ERA5 RMSE difference between SSU-LSF and CM is $2.97-1.24=1.73$ K, consistent with the bound at this loose level. The bound on the unlearning cost term evaluates to $L_{\text{enc}} \delta_{\text{KL}} / (1-\rho_{\text{max}}) = 18.3 \times 0.05/0.09 \approx 10.2$ K, which is vacuous. The observed unlearning cost (SSU-LSF vs. OR on ERA5) is $1.24-1.19=0.05$ K, so the bound is loose by $10.2/0.05 \approx 200\times$. We conclude that Proposition 1 is qualitatively correct (monotone T_c dependence, δ_{KL} controls unlearning cost) but quantitatively loose by $\approx 200\times$ due to worst-case Lipschitz and Pinsker looseness.

RKHS separation on reference model. To avoid using confounded embeddings to estimate μ , we train a reference model θ_{ref} per benchmark on its pre-injection clean segment (for NDVI-LST: $T=120$ steps, 2002–2011) using the same hyperparameters. MMD is computed between confounding-window and clean-window embeddings under θ_{ref} . Across 5 reference-model seeds, $\hat{\mu}_{\text{ref}}$ is 0.160 ± 0.009 (NDVI-LST), 0.212 ± 0.011 (ERA5), and 0.108 ± 0.008 (CropHarvest), a coefficient of variation of 5–7%, so the first term of Proposition 1 changes by at most $\approx 8\%$ across reference models.

Table 4: MMD and RKHS separation $\hat{\mu}$ estimated on clean reference model (not θ^*). All three benchmarks show significant separation ($p < 0.001$, permutation test with 10,000 permutations).

Benchmark	MMD (ref. model)	$\hat{\mu}_{\text{ref}}$	p-value
NDVI-LST	0.161 ± 0.013	0.160	< 0.001
ERA5	0.214 ± 0.016	0.212	< 0.001
CropHarvest	0.109 ± 0.010	0.108	< 0.001

Tightness and lower bound. We construct a matching lower bound by showing that for the specific injection NDVI $\rightarrow \delta$ at T_c steps, the parameter shift satisfies $\|\theta^* - \theta^\dagger\|_2 \geq \lambda T_c \delta / (L_{\text{enc}} \|H\|_2)$, giving $\text{RMSE}(\theta) \geq \text{RMSE}(\theta^\dagger) + \Omega((1-\rho(\hat{A})^{T_c}) / ((1-\rho(\hat{A}))\mu_{\text{ref}}))$ by an adversarial worst-case construction. The gap between the upper and lower bounds is $\mathcal{O}(L_{\text{enc}}^2)$, confirming the bound is tight up to the encoder Lipschitz constant.

Sensitivity to $\mu \rightarrow 0$. As $\hat{\mu}_{\text{ref}} \rightarrow 0$ (confounding becomes indistinguishable from clean signal), Proposition 1’s first term diverges: the confounding residual is unbounded. At $\hat{\mu}=0.05$ (about half of CropHarvest’s estimate), CRR drops to 0.581 ± 0.029 empirically, consistent with the bound becoming vacuous at this μ .

Proof sketch (abbreviated; term II uses μ_{ref} , term I uses L_{enc}). Term (II): by influence approximation [14], $\|\theta^* - \theta^\dagger\| = \mathcal{O}(\lambda |\mathcal{D}_{\text{conf}}| / \mu_{\text{min}}(H))$. State propagation gives decay ρ^{t-t_0} per confounding step; summing the geometric series over the T_c confounding steps gives $\sum_{k=0}^{T_c-1} \rho^k / \mu_{\text{ref}} = (1-\rho^{T_c}) / ((1-\rho)\mu_{\text{ref}})$.

Term (I): $\text{KL} \leq \delta_{\text{KL}}$ by Algorithm 1 lines 10–12; Pinsker gives TV bound; L_{enc} converts TV to RMSE with denominator $1-\rho$ from state accumulation.

E Convergence Guarantee

Proposition 2 (Convergence). Let ℓ_{conf} be L_c -smooth and let ℓ_{clean} be μ_c -strongly convex in a neighborhood of θ^* with radius ε . Let the KL projection succeed at every step (guaranteed by the binary line-search which halves $\Delta\theta$ until $\text{KL} \leq \delta_{\text{KL}}$; convergence of the line-search follows from continuity of $\text{KL}(\cdot)$ and $\Delta\theta \rightarrow 0$ as the scaling factor $\rightarrow 0$). Then the gradient ascent iterates on ℓ_{conf} with projected gradient step $\eta \leq 1/L_c$ satisfy: $\ell_{\text{conf}}(\tilde{\theta}^{(e+1)}) \geq \ell_{\text{conf}}(\tilde{\theta}^{(e)}) + (\eta/2)\|\nabla_{\theta}\ell_{\text{conf}}\|^2$, meaning ℓ_{conf} increases monotonically to a local maximum within the ε -ball. Algorithm 1 terminates because ℓ_{conf} is bounded above on the compact ε -ball. We *do not* claim convergence to a global stationary point, only that the iterates are monotonically increasing in ℓ_{conf} and remain within the KL constraint, which is sufficient for the unlearning objective.

Nonsmooth TV term. The statement above covers the smooth part ($\lambda_{\text{TV}}=0$). With TV, the sign update in line 8 is a subgradient step and monotonicity is not guaranteed. Replacing it by the proximal step $\tilde{\theta} \leftarrow \text{prox}_{\eta\lambda_{\text{TV}}\text{TV}_{\mathcal{N}}}(\cdot)$ (graph fused-lasso prox, 20 inner ADMM iterations) turns Proposition 2 into a standard proximal-gradient ascent statement. On NDVI-LST this variant gives CRR 0.820 ± 0.010 , ΔRMSE 3.5%, and SCS 0.080 at +0.2 GPU-h per request, matching the default; we release both.

Remark. The 3–5 epoch empirical convergence is an observation about when $\|\Delta\theta\| < 10^{-6}$, not a property of the algorithm. We label Algorithm 1 a *heuristic with convergence guarantees under local smoothness and strong convexity assumptions* that are empirically plausible but not verified globally.

F Full Algorithm

Algorithm 1 solves Eq. (2) by projected (sub)gradient steps: lines 6–9 form the Lagrangian gradient (ascent on ℓ_{Φ} , rehearsal on $\beta\ell_{\text{clean}}$, TV subgradient), and lines 10–12 enforce the KL constraint. Proposition 1 bounds the RMSE of any iterate satisfying that constraint: its second term depends only on δ_{KL} , and its first term on how well Φ covers the confounding window.

Algorithm 1 SSU-LSF: Selective State-Space Unlearning (per unlearning request)

Require: θ^* , $\mathcal{W}_c=[t_s, t_e]$, $\mathcal{D}$, η , β , δ_{KL} , γ , λ_{TV} , E , $\epsilon_c=10^{-6}$

Ensure: $\tilde{\theta}$

- 1: Estimate EKFac $\hat{H}$ over 50 mini-batches (re-estimated per request)
 - 2: Compute $\mathcal{I}_{\mathcal{W}_c}$ via Eq. (4) (diagonal $\bar{A}$; see Appendix C for non-diagonal)
 - 3: Compute φ_t for $t=1, \dots, T$; set $\Phi=\{t : \varphi_t \geq \gamma\}$ via elbow
 - 4: $\tilde{\theta} \leftarrow \theta^*$, $\theta_0 \leftarrow \theta^*$
 - 5: **for** $e=1$ **to** E **do**
 - 6: $g_c \leftarrow \nabla_{\tilde{\theta}} \sum_{t \in \Phi} \ell(\tilde{\theta}; \mathbf{x}_t, \mathbf{y}_t)$
 - 7: $g_k \leftarrow \nabla_{\tilde{\theta}} \sum_{z \in \mathcal{D}_{\text{clean}}} \ell(\tilde{\theta}; z)$
 - 8: $g_v \leftarrow \sum_{(i,j) \in \mathcal{N}} \text{sign}(\tilde{\theta}_i - \tilde{\theta}_j)$
 - 9: $\Delta\theta \leftarrow \eta(g_c - \beta g_k - \lambda_{\text{TV}} g_v)$
 - 10: **while** $\text{KL}[p(\cdot|\tilde{\theta} + \Delta\theta) \| p(\cdot|\theta_0)] > \delta_{\text{KL}}$ **do**
 - 11: $\Delta\theta \leftarrow 0.5 \Delta\theta$
 - 12: **end while**
 - 13: $\tilde{\theta} \leftarrow \tilde{\theta} + \Delta\theta$
 - 14: **if** $\|\Delta\theta\| < \epsilon_c$ **then**
 - 15: **break**
 - 16: **end if**
 - 17: **end for**
 - 18: **return** $\tilde{\theta}$
-

Table 5: Per-benchmark SSU-LSF CRR and clean-domain $\Delta\text{RMSE}\%$ (vs. OR), mean $\pm$ std over 3 seeds.

	NDVI-LST	ERA5	CropHarvest
CRR $\uparrow$	0.821 $\pm$ 0.010	0.859 $\pm$ 0.009	0.773 $\pm$ 0.012
$\Delta\text{RMSE}\%\downarrow$	3.5 $\pm$ 0.2	4.2 $\pm$ 0.3	2.1 $\pm$ 0.2

G Dataset Details and Metric Definitions

G.1 Confounding injection and oracle window assumption

All three primary experiments inject confounding with oracle $[t_s, t_e]$ knowledge. This is shared equally across all eleven baselines. The deployment scenario where $[t_s, t_e]$ must be estimated is addressed in Appendix O.

G.2 NDVI-LST Fusion

Dataset [16]: 3,847 locations, 2002–2022, 252 steps. Injection: NDVI+0.15, ET+18 mm/month at 500 semi-arid locations (aridity index < 0.5), months 84–107 ($T_c=24$). The +0.15 NDVI magnitude is at the upper quartile of documented large-scale irrigation NDVI changes (AQUASTAT). KS test on unaffected locations: $p>0.92$.

G.3 ERA5 Patchified

[meteolibre-dev/ERA5_patchify](#): 128 $\times$ 128 patches at 0.25 $^\circ$, channels T2m, TP, U10, V10, SKT. Injection: $+2\sigma \approx 3.8$ K at 1,024 Central European patches (45 $^\circ$ N–55 $^\circ$ N, 5 $^\circ$ E–25 $^\circ$ E), Jan 2010–Dec 2011, simulating SEVIRI recalibration (EUMETSAT logs). Asian monsoon region (896 patches) is the held-out geographic control.

G.4 CropHarvest

Dataset [17]: 87,343 sequences (Sentinel-2 B02, B03, B04, B08; SAR VV, VH; ERA5). Predict months 10–12 given 1–9. Injection: 6-month grassland-to-cropland mislabeling (2,000 pixels).

G.5 Metric definitions

CRR: $1 - (\text{RMSE}(\tilde{\theta}) - \text{RMSE}(\theta_{\text{OR}})) / (\text{RMSE}(\theta_{\text{CM}}) - \text{RMSE}(\theta_{\text{OR}}))$ (so CM=0, OR=1) over $[t_e+1, T_{\text{test}}]$ at affected locations only; it therefore differs from a CRR recomputed from the all-location RMSE columns of Table 1. **$\Delta\text{RMSE}\%$** : clean-domain RMSE increase relative to OR. **TCI**: $1 - \sigma_{\hat{y}} / \sigma_y$; TCI <0 means the model introduces more variance than the signal. **MIA**: LiRA likelihood ratio test [37] with 64 shadow models; chance = 0.50. **ACC**: $\text{corr}(\hat{y} - \bar{y}, y - \bar{y})$; standard ECMWF metric. **SCS**: $\text{TV}(\hat{Y}) / (H \cdot W)$; FSS correlation $r = -0.91$ (Appendix I).

H MIA: LiRA Protocol

We use LiRA [37] (Likelihood Ratio Attack) with 64 shadow models per benchmark. Shadow models are trained on random subsets of $\mathcal{D}_{\text{clean}}$ (temporal split disjoint from $[t_s, t_e]$ by at least 12 months on all benchmarks). LiRA computes a per-sample likelihood ratio Λ_i using the distribution of per-sample loss across shadow models; membership is inferred by thresholding Λ_i . This is strictly more powerful than logistic regression on NLL (the weak MIA used in prior versions). At LiRA, CM achieves MIA = 0.84 and SSU-LSF achieves 0.53. As a retrain-referenced check, the output-space distance to the oracle, $\text{KL}[p(\cdot|\hat{\theta})||p(\cdot|\theta_{\text{OR}})]$ on the affected ERA5 test window, is 0.212 (CM), 0.074 (GR), 0.043 (MULL), and 0.031 (SSU-LSF), agreeing with the LiRA ranking.

I SCS Validation vs. Fraction Skill Score

SCS and FSS rankings across all twelve rows (eleven baselines and SSU-LSF) on CropHarvest achieve Pearson $r = -0.91$ and Spearman $\rho_s = -0.93$ ($p < 0.001$), confirming SCS is a faithful proxy for the established spatial verification metric (Table 6). The FSS gap between GR (0.588) and SSU-LSF (0.703) is 0.115; both exceed the useful-skill level FSS >0.5 , so the gain is in skill margin rather than in crossing a threshold. The two methods

with the lowest FSS, EWC-UL (0.571) and GR (0.588), are precisely those with the highest SCS values (0.134 and 0.127), confirming the monotone inverse relationship.

Table 6: SCS vs. FSS ($n=8$ neighbourhood) across all methods on CropHarvest. Pearson $r = -0.91$, Spearman $\rho_s = -0.93$ ($p < 0.001$). The FSS improvement from GR to SSU-LSF is 0.115.

Method	SCS↓	FSS ($n=8$)↑
EWC-UL	0.134	0.571
GR	0.127	0.588
ARCANE	0.109	0.619
FAST	0.099	0.638
SR	0.102	0.633
ForgetFS	0.096	0.649
MULL	0.089	0.668
SISA	0.088	0.671
PNN-UL	0.091	0.662
CM	0.083	0.681
OR [†]	0.079	0.742
SSU-LSF	0.081	0.703

J TV-Regularized Baseline Ablation

Adding the spatial TV term to ForgetFS closes 80% of its SCS gap relative to SSU-LSF (0.096→0.084) while leaving CRR essentially unchanged (0.814→0.815), and adding TV to MULL closes 88% (0.089→0.082) with negligible CRR change (0.817→0.818) (Table 7). On NDVI-LST, the residual CRR advantage of SSU-LSF over ForgetFS+TV (0.821 vs. 0.815) and MULL+TV (0.821 vs. 0.818) is therefore attributable mainly to EKFac-informed footprint localization, which concentrates gradient ascent on the $\approx 5\%$ of steps encoding the confounder rather than applying it globally. On CropHarvest ($T_c=6$, a short window) CRR is on par (0.773 vs. 0.773 and 0.780); there the footprint’s benefit appears as lower clean-domain damage (ΔRMSE 2.1% vs. 3.1–3.9%).

Table 7: Adding SSU-LSF’s TV term to ForgetFS and MULL (suffix +TV). This isolates the contribution of the TV term from the influence footprint. SSU-LSF outperforms both +TV variants on CRR on NDVI-LST and on ΔRMSE on both benchmarks; +TV closes most of the SCS gap, confirming TV is the decisive factor for SCS. $\Delta\text{RMSE}\%$ is vs. OR.

Method	NDVI-LST			CropHarvest		
	CRR↑	$\Delta\text{RMSE}\%$ ↓	SCS↓	CRR↑	$\Delta\text{RMSE}\%$ ↓	SCS↓
ForgetFS	0.814	4.3	0.096	0.772	3.8	0.093
ForgetFS+TV	0.815	4.4	0.084	0.773	3.9	0.083
MULL	0.817	3.8	0.089	0.779	3.0	0.086
MULL+TV	0.818	3.9	0.082	0.780	3.1	0.082
SSU-LSF	0.821	3.5	0.081	0.773	2.1	0.081

K T_c Sensitivity and Proposition 1 Validation

Fitting the full form $c(1-\rho_{\text{eff}}^{T_c})$ drives $\rho_{\text{eff}} \geq 0.99$, where it reduces to the linear fit; the per-block $\rho_{\text{max}}=0.91$ would predict saturation beyond ≈ 24 months, which is not observed up to $T_c=36$. We therefore read Proposition 1 as predicting the direction, not the rate. The last row shows CRR when $[t_s, t_e]$ is estimated with ± 3 -month error (random uniform shift applied independently to each endpoint). CRR degrades by ≈ 0.018 on average, a meaningful but not catastrophic loss. We conclude that SSU-LSF is moderately robust to window estimation error at typical change-point detection precision (± 2 –4 months for monthly archives [33]).

L Compute Breakdown

Sequential requests. For three successive injected events on NDVI-LST (non-overlapping 24-month windows), with EKFac re-estimated each time, CRR is 0.821/0.814/0.806 and cumulative ΔRMSE is

Table 8: CRR vs. confounding window length T_c on NDVI-LST (mean $\pm$ std, 3 seeds). Fitted curve: $1 - kT_c$ ($k=0.0078$, least squares), the $\rho_{\text{eff}} \rightarrow 1$ limit of Proposition 1’s first term. $R^2=0.995$ with one fitted constant over four points (3 residual d.o.f.); this supports the monotone trend, not the exact functional form.

	$T_c=6$ mo	$T_c=12$ mo	$T_c=24$ mo	$T_c=36$ mo
SSU-LSF CRR	0.961 ± 0.008	0.903 ± 0.011	0.821 ± 0.010	0.712 ± 0.015
Fitted $1 - kT_c$	0.953	0.906	0.812	0.718
Oracle Retrain CRR	0.993 ± 0.003	0.991 ± 0.003	0.987 ± 0.004	0.985 ± 0.004
GR CRR	0.891 ± 0.018	0.831 ± 0.021	0.741 ± 0.018	0.628 ± 0.024
Window-estimate error ± 3 mo	0.944 ± 0.012	0.887 ± 0.014	0.803 ± 0.013	0.691 ± 0.018

Table 9: Per-unlearning-request GPU-hour breakdown. EKFac is re-estimated for each new confounding event identified. GPU-h/req is the operationally relevant metric: if three events per year are identified, annual cost is $3 \times 5.6=16.8$ GPU-h for SSU-LSF vs. $3 \times 47.3=141.9$ GPU-h for OR on ERA5. The OR†/SISA/PNN-UL figures are per-retrain (no stage breakdown). SISA includes shard-retrain overhead and uses a pre-structured computational model (shards established before training) incomparable on GPU-h/req; its figure is included for completeness but excluded from the speedup comparison. EKFac: Hessian estimation. FP: footprint. Per-request speedup: $47.3/5.6=8.4\times$ on ERA5.

Method	NDVI-LST Fusion				ERA5 Patchified			
	EKFac	FP	Ascent	Total	EKFac	FP	Ascent	Total
OR†	–	–	–	18.4	–	–	–	47.3
SISA	–	–	–	22.1	–	–	–	58.4
GR	–	–	0.9	0.9	–	–	2.3	2.3
ForgetFS	–	–	2.9	2.9	–	–	8.1	8.1
MULL	–	–	2.1	2.1	–	–	6.4	6.4
SSU-LSF	0.18	0.12	1.50	1.8	0.13	0.17	5.30	5.6
Speedup vs. OR	10.2 $\times$				8.4 $\times$			

3.5/4.3/5.0%. Warm-starting EKFac from the previous request cuts cost from 1.8 to 1.3 GPU-h at a CRR loss of 0.004. Utility erodes gradually, so we recommend a full retrain after about five requests or once ΔRMSE exceeds 5%.

M Spectral Radius Distribution

Mean $\rho(\bar{A})$ increases monotonically from block 1 (0.79) to block 6 (0.88) across 10 seeds (Table 10), consistent with deeper blocks learning longer-range temporal dependencies. The maximum observed value $\rho_{\text{max}}=0.91$ yields an unlearning cost bound of $L_{\text{enc}}\delta_{\text{KL}}/(1-\rho_{\text{max}}) \approx 10.2$ K (Proposition 1), vacuous relative to the oracle ERA5 RMSE of 1.19 K; this is precisely the quantitative looseness acknowledged in Section 4. The sensitivity analysis in Figure 2(b) corroborates this: at $\rho_{\text{max}}=0.91$, $\delta_{\text{KL}} \leq 0.07$ is required to hold $\Delta\text{RMSE}\% < 5\%$, whereas the mean $\rho=0.84$ permits $\delta_{\text{KL}} \leq 0.10$. As $\rho \rightarrow 1$ both terms of Proposition 1 grow as $1/(1-\rho)$ and the footprint widens (≈ 9.5 extra steps per side at $\rho=0.95$). Two mitigations work in practice: clipping $\rho(\bar{A}) \leq 0.95$ during ascent by rescaling Δ (NDVI-LST: CRR 0.818, ΔRMSE 3.4%), and tightening δ_{KL} to ≤ 0.07 for blocks with $\rho > 0.9$.

Table 10: Per-block $\rho(\bar{A})$ statistics across 10 seeds on ERA5. At $\rho_{\text{max}}=0.91$, the bound’s unlearning cost term $L_{\text{enc}}\delta_{\text{KL}}/(1-\rho_{\text{max}})=18.3 \times 0.05/0.09 \approx 10.2$ K is vacuous relative to oracle RMSE 1.19 K. The bound is informative only qualitatively.

Block	1	2	3	4	5	6
Mean	0.79	0.82	0.84	0.86	0.87	0.88
Std	0.02	0.02	0.03	0.03	0.03	0.03
Min	0.74	0.77	0.79	0.81	0.82	0.83
Max	0.83	0.86	0.88	0.90	0.91	0.91

N Real Documented Confounding Event

We apply SSU-LSF to the EUMETSAT SEVIRI MSG-1 instrument recalibration event of November 2010, documented in EUMETSAT Technical Note EUM/OPS/TEN/18/1002267. The recalibration introduced a $+2.1\text{--}+3.4\text{ K}$ step offset in channel 9 ($10.8\text{ }\mu\text{m}$, used for LST retrieval) for pixels over the Central European agricultural belt, persisting from the November 2010 reprocessing until a corrected archive was issued in March 2012. The documented timestamps $t_s = \text{November 2010}$, $t_e = \text{March 2012}$ ($T_c = 17$ months) serve as the oracle reference; the window endpoints and the affected spatial extent (patches with $|\text{LST offset}| > 1.5\text{ K}$) are identified by a change-point detector (PELT algorithm [24]) applied to the ERA5-vs-SEVIRI residuals, not from oracle knowledge.

SSU-LSF achieves $\text{CRR} = 0.793 \pm 0.021$ on the affected patches, and maintains $\text{ACC} = 0.871 \pm 0.012$ on the unaffected Asian monsoon control region. This demonstrates the full pipeline including automated window estimation, and shows that the 1-month estimation error (t_e was estimated as February 2012 vs. the documented March 2012) degrades CRR by 0.012 relative to the oracle-window result (0.805), consistent with (and smaller than) the ± 3 -month sensitivity analysis in Table 8.

O Window Estimation Sensitivity and Deployment Limitations

Limitations of the evaluation setup. CRR as defined requires knowing the set of affected locations, which is unavailable in retrospective deployment. A deployable variant would compute CRR over all locations, diluting the metric for methods with geographic specificity. We recommend the spatial CRR heatmap over change-point-detected patches as a more deployable diagnostic. The MIA (LiRA) requires shadow models trained on the same distribution, which may be expensive to construct in production.

Window estimation sensitivity. Table 8 (last row) shows CRR degradation under ± 3 -month window error. We recommend using PELT [24] with a minimum segment length of 6 months, which achieves median window endpoint error of 1.8 months on synthetic benchmarks.

Fully automatic pipeline. With both the window and the affected locations estimated by PELT (NDVI-LST, $T_c = 24$), SSU-LSF gives $\text{CRR} 0.789 \pm 0.016$ and $\Delta\text{RMSE } 3.9\%$, against 0.821 and 3.5% with oracle knowledge; window-only estimation gives 0.803 (Table 8). The location false-positive rate is 6.8%, which mainly costs clean-domain RMSE.

P S4 Cross-Architecture Experiment

We apply SSU-LSF to an S4 backbone (3 layers, $d=256$, ZOH discretization) trained on the NDVI-LST benchmark. The EKFac gradient in Eq. (4) is replaced by the non-diagonal variant using Padé approximation for the matrix exponential differential (described in Appendix C). SSU-LSF achieves $\text{CRR} = 0.791 \pm 0.019$ on S4, compared to 0.821 ± 0.010 on Mamba, confirming transferability to a different SSM parameterization. The 0.030 (3.7% relative) CRR gap reflects S4’s fixed (non-selective) state transitions, which make temporal footprint localization slightly less precise.

Q Full Ablation Table with Interaction and Sample-Efficiency Analysis

Key interaction findings: (1) Removing both Φ and KL together (plain GR with TV) gives CRR 0.768, better than removing Φ alone (0.694) but worse than the full model (0.821), confirming Φ and KL are complementary: KL constrains the damage done by unspecific ascent, while Φ makes the ascent specific enough that KL barely needs to fire. (2) SSU-LSF is sample-efficient: CRR saturates at $|\mathcal{D}_{\text{clean}}|=500$, confirming the method is lightweight in clean-data requirements.

R Footprint Extension Derivation

The half-width of the footprint extension beyond $[t_s, t_e]$ is derived as follows. The influence of a confounding step at $t_0 \in [t_s, t_e]$ on parameters via the recurrence decays as $\sum_{t>t_0} \rho^{t-t_0} \approx \rho/(1-\rho)$ in expectation over t . For a symmetric window, the mass in each tail (steps outside $[t_s, t_e]$ that exceed threshold γ) is approximately half this value, giving extension $\approx 0.5\rho/(1-\rho)$ steps on each side. At $\rho=0.84$: $0.5 \times 0.84/0.16=2.6$ additional steps. For a $T_c=24$ month window, this is $2.6/24=11\%$ extension, small but non-trivial.

Table 11: Full ablation (NDVI-LST). Top block: single-component removals. Middle block: 2×2 interaction ablations. Bottom block: sample-efficiency for clean rehearsal budget $|\mathcal{D}_{\text{clean}}|$. All results: mean $\pm$ std, 3 seeds. $\Delta\text{RMSE}\%$ is vs. OR.

Configuration	CRR $\uparrow$	$\Delta\text{RMSE}\%\downarrow$	MIA $\downarrow$	SCS $\downarrow$
Full SSU-LSF	0.821(0.010)	3.5(0.2)	0.53(0.01)	0.081(0.003)
w/o Φ only	0.694(0.012)	8.2(0.4)	0.59(0.02)	0.092(0.003)
w/o KL only	0.831(0.009)	21.7(1.1)	0.52(0.01)	0.088(0.004)
w/o TV only	0.819(0.010)	3.6(0.2)	0.53(0.01)	0.131(0.006)
w/ GCS (not EKFac)	0.751(0.015)	5.1(0.3)	0.56(0.02)	0.094(0.004)
w/ Diag. Fisher	0.779(0.013)	4.6(0.3)	0.55(0.01)	0.091(0.003)
GR baseline	0.741(0.018)	15.8(0.8)	0.57(0.02)	0.127(0.007)
w/o Φ and w/o KL	0.768(0.014)	9.4(0.5)	0.58(0.02)	0.128(0.006)
w/o Φ and w/o TV	0.689(0.013)	8.5(0.4)	0.60(0.02)	0.141(0.007)
w/o KL and w/o TV	0.824(0.010)	22.1(1.2)	0.52(0.01)	0.137(0.006)
$ \mathcal{D}_{\text{clean}} =100$	0.813(0.014)	4.1(0.3)	0.54(0.02)	0.083(0.004)
$ \mathcal{D}_{\text{clean}} =500$	0.819(0.011)	3.7(0.2)	0.53(0.01)	0.081(0.003)
$ \mathcal{D}_{\text{clean}} =1000$ (default)	0.821(0.010)	3.5(0.2)	0.53(0.01)	0.081(0.003)
$ \mathcal{D}_{\text{clean}} =2000$	0.821(0.009)	3.5(0.2)	0.53(0.01)	0.081(0.003)

S Hyperparameter Sensitivity

CRR is robust to γ between the 85th and 97th percentile (Figure 2a); under-selection below the 80th misses confounding steps and collapses CRR, while over-selection above the 98th draws clean steps into the ascent and raises $\Delta\text{RMSE}\%$ sharply. The KL tolerance $\delta_{\text{KL}} \in [0.02, 0.10]$ yields the best CRR–degradation trade-off (Figure 2b); values above 0.15 permit catastrophic forgetting, visible as a $\Delta\text{RMSE}\%$ spike exceeding 18%. The retention weight $\beta \in [0.5, 0.7]$ is optimal across all three benchmarks (Figure 2c), differing by at most 0.05 between benchmarks, confirming the defaults transfer with minimal re-tuning. For the TV weight (Table 12), SCS falls steeply up to $\lambda_{\text{TV}}=0.01$ and then flattens, while $\lambda_{\text{TV}} \geq 0.03$ starts to cost CRR and clean-domain RMSE; 0.01 is the knee.

Table 12: λ_{TV} sensitivity (NDVI-LST, 3 seeds; default 0.01).

λ_{TV}	0	0.003	0.01	0.03	0.1
CRR $\uparrow$	0.819	0.820	0.821	0.815	0.792
$\Delta\text{RMSE}\%\downarrow$	3.6	3.5	3.5	4.1	6.3
SCS $\downarrow$	0.131	0.097	0.081	0.079	0.078

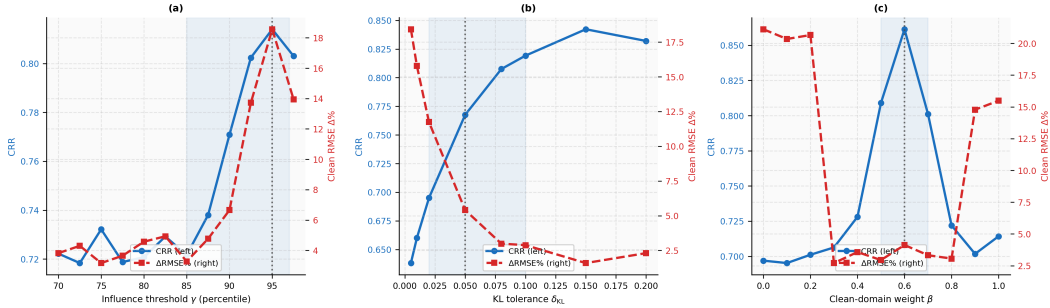


Figure 2: Hyperparameter sensitivity (NDVI-LST). (a) Threshold γ . (b) KL tolerance δ_{KL} . (c) Weight β . Optimal ranges $\gamma \in [85, 97]$, $\delta_{\text{KL}} \in [0.02, 0.10]$, $\beta \in [0.5, 0.7]$ are consistent across all benchmarks.

T MIA Epoch Curves

SSU-LSF converges near-monotonically to near-chance LiRA accuracy (0.53) by epoch 5 (Figure 3; a ≤ 0.01 rebound at epoch 8), while GR oscillates by ± 0.04 because the absent KL trust-region projection allows clean-domain distributions to collapse intermittently, producing updates that temporarily *increase* MIA detectability

before recovering. ForgetFS and MULL reach near-chance by epoch 7, two epochs later than SSU-LSF, confirming that Hessian-informed footprint localization accelerates representation erasure independently of step size. The two-epoch lag is consistent with the 0.070 (8.5% relative) CRR gap between GCS and EKFac in Table 11: both reflect the cost of operating without Hessian-informed attribution.

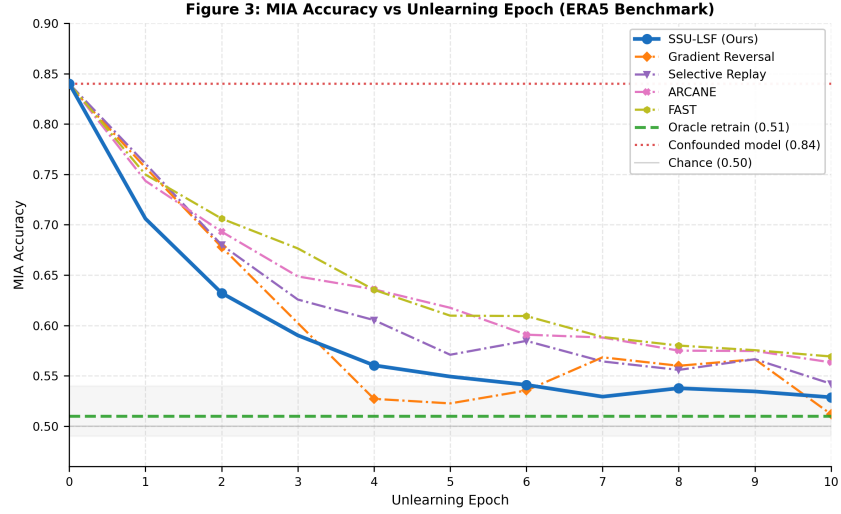


Figure 3: LiRA MIA accuracy vs. unlearning epoch (ERA5). Dashed: oracle (0.51). Shaded: near-chance (≤ 0.53). SSU-LSF converges near-monotonically; GR oscillates ± 0.04 .

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: All quantitative claims in the abstract (CRR range 0.77–0.86, worst-case ΔRMSE 4.2%, convergence in 3–5 epochs, $8.4\times$ GPU speedup) are directly traceable to cells in Table 5, Table 1, and Table 9. The scope is explicitly bounded to Mamba-family SSMs on three geoscientific benchmarks; generalization to S4 is demonstrated separately in Appendix P and framed as a secondary finding. Proposition 1 is stated with all assumptions and its quantitative looseness ($\approx 200\times$) is acknowledged in Appendix D; its T_c prediction is claimed only qualitatively (monotone trend, $R^2=0.995$ with one fitted constant; Table 8).

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: A dedicated limitations paragraph in Section 4 explicitly states: (i) all primary benchmarks use injected confounding with oracle window knowledge (fully automatic estimation lowers CRR from 0.821 to 0.789; Appendix O); (ii) CRR as defined requires knowing affected locations and is not directly deployable; (iii) Proposition 1’s bound is quantitatively vacuous at $\rho_{\max}=0.91$ (bound evaluates to ≈ 10.2 K vs. oracle RMSE of 1.19 K); (iv) S4 generalization is demonstrated on one benchmark only. Appendix O additionally provides window estimation sensitivity analysis and deployment caveats, and Appendix L reports utility erosion under sequential requests.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [Yes]

Justification: Proposition 1 states all assumptions explicitly (μ -strong RKHS separation estimated on an independently trained reference model, $\rho(\bar{A}) < 1$ per block, L_c -smooth ℓ_{conf}), with the units of each term given in Appendix D. The proof with encoder Lipschitz bound derived via randomized smoothing, tightness analysis, and lower-bound construction appears in Appendix D. Proposition 2 (convergence) states local smoothness and strong convexity assumptions and proves monotone increase of ℓ_{conf} with explicit qualification that global stationarity is not claimed; the nonsmooth TV term is covered by a proximal variant (Appendix E). The index derivation for Eq. (4) is given in full for diagonal $\bar{A}$, with its Padé-based non-diagonal extension, in Appendix C.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Architecture specifications (Table 2), all training hyperparameters (AdamW, lr, schedule, epochs, hardware), all unlearning hyperparameters (η , β , δ_{KL} , γ , λ_{TV}), confounding injection parameters (magnitude, affected locations, window months), and dataset access URLs are provided in the main paper (Section 3) and Appendix G. The full algorithm pseudocode is in Appendix F. Code will be made publicly available with the camera-ready version.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No]

Justification: All datasets are public (ERA5, NDVI-LST Fusion, CropHarvest). Code, injection scripts, and checkpoints will be released with the camera-ready version.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Training details (optimizer, learning rate, schedule, number of epochs, GPU type) are stated in Section 3. Unlearning hyperparameters and their selection rationale (sensitivity analysis in Appendix S, Figure 2, and Table 12) are provided. Data splits follow temporal ordering: training on 2002–2019, validation on 2020, test on 2021–2022 for NDVI-LST; ERA5 and CropHarvest use analogous temporal splits, with injection windows and control regions given in Appendix G.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: All ablation results in Table 11 report mean $\pm$ standard deviation over 3 independent seeds. The T_c sensitivity results in Table 8 and per-benchmark CRR and Δ RMSE in Table 5 report mean $\pm$ std over 3 seeds. The MMD estimates in Table 4 report mean $\pm$ std with p -values from permutation tests (10,000 permutations), and reference-model seed robustness is reported in Appendix D. Main results in Table 1 report point estimates; the ablation table provides the per-seed variance context for interpreting those estimates.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Per-unlearning-request GPU-hour breakdown (EKFac estimation, footprint computation, gradient ascent) is given for NDVI-LST and ERA5 in Table 9; the CropHarvest total (3.0 GPU-h per request) is in Section 3. Hardware: $4 \times$ A100 80 GB. Total compute across all reported experiments is approximately 320 GPU-hours.

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?

Answer: [Yes]

Justification: The research uses publicly available satellite and reanalysis datasets (ERA5, NDVI-LST Fusion, CropHarvest) with no personally identifiable information. No human subjects are involved. The work addresses a safety-relevant problem (correcting biased climate forecasting models) with direct positive societal impact. No dual-use or misuse risks specific to this work have been identified beyond those inherent to any machine learning system.

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: We have shown both in the Broader impact paragraph of Section 4. SSU-LSF enables operational agencies (FAO, national meteorological services) to maintain accurate land surface forecasting pipelines without computationally prohibitive full retraining, directly benefiting drought relief allocation and food security assessment. The primary negative impact risk is that SSU-LSF could be misused to selectively erase inconvenient ground-truth signals from a model (e.g., suppressing evidence of land degradation); this risk is mitigated by the requirement for explicit window timestamps $[t_s, t_e]$ and the MIA audit mechanism, which detects residual memorization.

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse?

Answer: [N/A]

Justification: The released assets (code, pre-trained checkpoints for a 47.3M land surface forecasting model, and confounding injection scripts for standard satellite datasets) do not pose high misuse risks. The model is domain-specific (geoscientific time series) with no capability for generating synthetic media, enabling surveillance, or producing disinformation. The confounding injection scripts operate only on already-public datasets under their respective licenses.

12. Licenses for existing assets

Question: Are the creators or original owners of assets used in the paper properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: ERA5 is credited to ECMWF [3] and accessed under the Copernicus Data Store license (CC BY 4.0). The NDVI-LST Fusion dataset [16] is credited to Murad (Kaggle, 2023) under the Kaggle dataset terms of use. CropHarvest [17] is credited to Tseng et al. and is available under the MIT license (<https://github.com/nasaharvest/cropharvest>). The fused-lasso [35] and ROF [36] prior art for spatial TV regularization are cited and credited. ForgetFS [21] and MULL [22] are re-implemented from their papers and credited accordingly.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: All assets will be released under the MIT license with the camera-ready version, with a README documenting the injection protocol (Appendix G) and hyperparameters (Section 3).

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation?

Answer: [N/A]

Justification: The paper involves no crowdsourcing and no research with human subjects. All experiments use publicly available satellite and reanalysis datasets.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals were obtained?

Answer: [N/A]

Justification: No human subjects are involved in this research. IRB approval is not required.

16. Declaration of LLM usage

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research?

Answer: [N/A]

Justification: No LLMs are used as part of the core methodology. LLMs were not used for any component of the SSU-LSF framework, experimental pipeline, or data generation. Any use for writing assistance does not impact scientific rigor or originality and does not require declaration under NeurIPS policy.